Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 22)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 2 Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 1 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Do not write outside the box

Isotope analysis of ice cores .....

.....

Gas analysis of ice cores .....

.....

Radiocarbon dating .....

.....

Magnesium : calcium ratio in calcite .....

.....

10

0 2 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

.....

.....

.....

.....

0 2 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

.....

.....

Do not write outside the box

Do not write outside the box

0 2 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

10

0 3 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 3 . 2 Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

Do not write outside the box

0 3 . 3 Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

10

0 4 Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 4 . 1 Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

Do not write outside the box

.....

.....

.....

0 4 . 2 Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

.....

.....

.....

.....

0 4 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

.....

.....

.....

.....

0 4 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

.....

.....

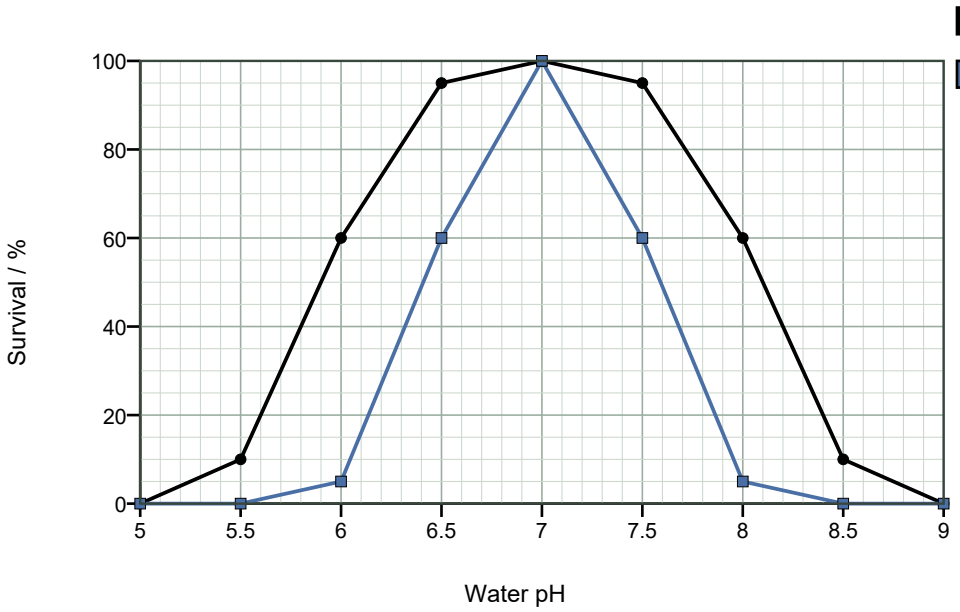
.....

.....

10

0 5 Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 5 . 1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 5 . 2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

.....

0 5 . 3 Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

.....

.....

.....

.....

0 5 . 4 Give **two** biotic factors that affect the distribution of a species.

[2 marks]

1 .....

.....

2 .....

.....

10

0 6 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 6 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

.....

.....

.....

.....

0 6 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

.....

.....

.....

.....

5

0 7 . 1 Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber .....

.....

Fuel .....

.....

Food .....

.....

Fibres .....

.....

Medicines .....

.....

0 7 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 7 . 3 Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

.....

.....

.....

.....

10

0 8 . 1 Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

.....

.....

.....

.....

.....

.....

.....

0 8 . 2 Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

.....

.....

0 8 . 3 Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

.....

.....

Do not write outside the box

Do not write outside the box

0 8 . 4 Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

10

0 9 Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 9 . 1 Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

Do not write outside the box

0 9 . 2 Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

0 9 . 3 Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

0 9 . 4 Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

10

1 0 . 1 Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

1 0 . 2 State what is meant by **rewilding**.

[1 mark]

10

Write an essay on one of the following topics.

1 1 . 1 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS